

Faculty of Engineering & Technology
Gurukul Kangri University , Haridwar
First Sessional Exam (2025-26)
Embedded Systems and IoT Applications
BET-C620

Time : 1 Hr.

M.M: 20

Section A : Attempt any Two question. Each question carry 5 marks.

Question : 1 Define Embedded System . Also discuss basic components required for design an embedded system.

Question : 2 What do you mean by Real Time Operation System (RTOS). How RTOS plays an important role in designing an Embedded System.

Question : 3 Define the following

- (a) Tri state Buffer
- (b) Watch Dog Timer

Section B : Attempt any one question. Each question carry 10 marks.

Question: 1 Discuss in detail the Design Life Cycle of an Embedded System.

Question : 2 How can you define the hardware and software partitioning and co-designing for an embedded system.

Course name: Embedded System & IOT Applications**Course code: BET-C620****Branch: CSE (VI sem)****Time: 1Hr****M.M. 20****Note:** *Attempt any two from section A and one from section B.****Section A (2*6)***

Q.No	Question statement	COs	BL
1	Explain Direct Memory Access (DMA) and its advantages in data transfer.	CO4	L2
2	Discuss the addressing modes of microcontroller 8051 with an example.	CO3	L5
3	Describe the I ² C serial communication protocols used in embedded systems.	CO1	L1
4	What is arbitration? Explain the role of arbiter in embedded system.	CO3	L3

Section B (1*8)

1	Describe TCON, TMOD & SCON registers of 8051 microcontroller.	CO1	L1
2	Write an assembly program to generate a unipolar square waveform of 1Khz frequency using the timer 0 of 8051 in mode 1. Assume the crystal oscillator Frequency 12 Mhz.	CO4	L5